

Module 2

Filesystem, Permissions & Environment

Learning objectives

- Read ls -l output (types, permissions, owner)
- Use chmod to make scripts executable safely
- Explain HOME, PATH, and which
- Fix common permission denied and command not found errors

Prerequisites

- Module 1 complete (cd, ls, pwd, basic files)

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[File types] --> B[Permissions]

B --> C[Ownership]

C --> D[Environment PATH]

D --> E[Dotfiles]

E --> F[Module 3]

Course context

- Simulators and tools depend on PATH
- Scripts in projects need chmod +x
- Dotfiles like .gitignore appear in every repo

Core topics

Self-check

```
$ ./scripts/module2.sh --check
```

Terminal: module_check

```
[[0;36m===== [[0m
[[0;36mModule 2: Environment self-check [[0m
[[0;36m===== [[0m

[[0;32m[OK] [[0m ls available
[[0;32m[OK] [[0m chmod available
[[0;32m[OK] [[0m which available
[[0;32m[OK] [[0m readlink available
[[0;32m[OK] [[0m HOME set: /home/yongfu
[[0;32m[OK] [[0m PATH set
[[0;32m[OK] [[0m module2 directory exists: /home/yongfu/proj/universal-verification-methodology/l
[[0;32m[OK] [[0m file_types example exists
[[0;32m[OK] [[0m permissions/hello.sh exists
[[0;32m[OK] [[0m file_types symlink (my_symlink) exists
[[0;32m[OK] [[0m environment/inspect_env.sh exists
[[0;32m[OK] [[0m path_and_script/local_cmd.sh exists and is executable

[[0;32m[OK] [[0m All required checks passed.
```

Make a script executable

```
chmod u+x hello.sh  
./hello.sh
```

ls -l and permissions

```
$ cd module2/examples/permissions && ls -l hello.sh config.txt
```

Terminal: permissions_ls

```
-rwxrwxrwx 1 yongfu yongfu 145 May 28 12:03 config.txt  
-rwxrwxrwx 1 yongfu yongfu 150 May 28 12:03 hello.sh
```

Finding commands on PATH

```
$ which bash && which git
```

Terminal: which_tools

```
/usr/bin/bash  
/usr/bin/git
```

Practice

Exercise scaffold

```
./scripts/module2.sh --scaffold
```

Exercises

- `chmod` on `~/unix_practice/hello.sh`
- `echo $PATH` and `which` for `git`
- `ls -la` to see dotfiles in `examples/dotfiles`

Summary & next steps

- File types and rwx permissions
- PATH and environment
- Next: Module 3 — processes & pipes
- Next: Module 3: Processes, Pipes & Redirection